

1. Summarizing Data

In this exercise, you will be working with the file **qb_stats.csv**. This file contains stats on NFL quarterbacks in the 2019 season. The data was obtained from <https://www.footballdb.com/statistics/nfl/player-stats/passing/2019/regular-season>. The data includes the following.

- player's name
- team
- number of games played
- pass attempts
- pass completions
- completion percentage
- number of yards
- yards per attempt
- touchdowns
- interceptions
- sacks
- quarterback rating

Use Python code to answer the following questions.

- Find the mean of each numerical column of data.
- Find the standard deviation of each numerical column of data.
- Create a histogram of the number of yards; label it appropriately.
- Create a boxplot of the number of touchdowns. Identify any outliers.

2. Calculating Probabilities from Data

In this exercise, you will be working with the file **survey_data.csv**. This data has a list of college students who were asked their hair color and eye color.

Use Python code to answer the following questions.

- Based off of this data, what is the probability a college student has brown hair?
- Based off of this data, what is the probability a college student has blue eyes?
- Based off of this data, what is the probability a college student has blue eyes given that they have brown hair?
- Based off of this data, what is the probability a college student has brown hair given that they have blue eyes?
- Do your results above indicate that college students having brown hair and blue eyes are independent of one another? Explain.
- Create a bar graph of the hair color and eye color of this group of students. Label the graphs appropriately.